For example, we agree with critics who note that the Black–Scholes model is *wrong*, in the sense that it makes predictions about option prices that are in some ways systematically inconsistent with the prices of options as repeatedly observed in various markets. However, we've found the Black–Scholes model to be useful in many contexts, as have a large number of analysts and traders. It's important to be familiar with its problems and pitfalls, and like most tools it can do damage if used improperly. But we recommend it as a tool of the trade that is quite useful in a number of contexts.

Analytical Scope (Applicability)

For our purposes, it's also useful for a model to have a broad scope, with applicability to a wide range of situations. For example, principal component analysis (PCA) has proven to be useful in a large number of applications, including interest rates, swap spreads, implied volatilities, and the prices of equities, grains, metals, energy, and other commodities. As with any powerful model, there is a cost to implementing PCA, but the applicability of the model once it has been built means that the benefits of the implementation tend to be well worth the costs.

Other statistical models with broad applicability are those that characterize the mean-reverting properties of various financial variables. Over considerable periods of time, persistent mean reversion has been observed in quite a large number of financial variables, including interest rates, curve slopes, butterfly spreads, term premiums, and implied volatilities. And in the commodity markets, mean reversion has been found in quite a number of spreads, such as those between gold and silver, corn and wheat, crack spreads in the energy complexes, and crush spreads in the soybean complex.

The ubiquity of mean-reverting behavior in financial markets implies that mean reversion models have a tremendous applicability. As a result, we consider them some of the more useful tools of a well-equipped relative value analyst, and we discuss them in some detail in this book.

Parsimony

From our perspective, it's also useful for a model to be parsimonious. As Einstein articulated in his 1933 lecture "On the Method of Theoretical Physics," "It can scarcely be denied that the supreme goal of all theory is to make irreducible basic elements as simple and as few as possible without having to surrender the adequate representation of a single datum of experience."

In our context, it's important to note the relative nature of the word "adequate." In most circumstances, there is an inevitable tradeoff between the parsimony of a model and its ability to represent experience. The goal of people developing models is to improve this tradeoff in various contexts. The goal of people using models is to select those models that offer the best tradeoff between costs and benefits in specific applications. And it's in that sense that we characterize the models in this book as being useful in the context of relative value analysis.

SUMMARY OF CONTENTS

Relative value analysis models can be divided into two categories: statistical and financial. Statistical models require no specific knowledge about the instrument that is being modeled and are hence universally applicable. For example, a mean reversion model only needs to know the time series, not whether the time series represents yields, swap spreads, or volatilities, nor what drives that time series.

Financial models, on the other hand, give insight into the specific driving forces and relationships of a particular instrument (and are therefore different for each instrument). For example, the specific knowledge that swap spreads versus LIBOR include the unsecured–secured basis can explain its statistical behavior.

While we present the models in two separate categories, comprehensive relative value analysis combines both. The successful relative value trader described above might first use statistical models to identify which instruments are rich and cheap relative to each other, and then apply financial models in order to gain insights into the reasons for that richness and cheapness, on which basis he can assess the likelihood for the richness and cheapness to correct. If he sees a sufficient probability for the spread position to be an attractive trade, he can then use statistical models again to calculate, among others, the appropriate hedge ratios and the expected holding horizon.

Statistical Models

The two types of statistical models presented here are designed to capture two of the most useful statistical properties frequently observed in the fixed income markets: the tendency for many spreads to revert toward their longer-run means over time and the tendency for many variables to increase and decrease together. The first chapters model these properties separately, while the last chapter models the two properties simultaneously.

Mean Reversion

Many financial spreads exhibit a persistent tendency to revert toward their means, providing a potential source of return predictability. In this chapter, we discuss stochastic processes that are useful in modeling this mean reversion, and we present ways in which data can be used to estimate the parameters of these processes. Once the parameters have been estimated, we can calculate the half-life of a process and make probabilistic statements about the value of the spread at various points in the future.

We also present the concept of a first passage time and show ways to calculate probabilities for first passage times. Once we have these first passage time densities, we can provide probabilistic answers to some of the more perplexing questions that are typical on a trading desk. *Over what time period should I expect this trade to perform? What sort of return target is reasonable over the next month? How likely am I to hit a stop-loss if placed at this level?* First passage time densities can provide probabilistic answers to these questions, and we discuss practical ways in which they can be implemented in a trading environment.

Principal Component Analysis

Many large data sets in finance appear to be driven by a smaller number of factors, and the ability to reduce the dimensionality of these data sets by projecting them onto these factors is a very useful method for analyzing and identifying relative value opportunities. In this chapter we discuss PCA in some detail. We address not only the mathematics of the approach but also the practicalities involved in applying PCA in real-world applications, including trading the underlying factors and hedging the factor risk when trading specific securities.

Multivariate Mean Reversion

Chapter 2 discusses ways to model univariate series that exhibit mean reversion and Chapter 3 ways to model multivariate series that exhibit correlation. In this chapter, we discuss the multivariate Ornstein–Uhlenbeck process as a way to model multivariate series that exhibit both mean reversion and correlation. This combined perspective is capable of capturing a richer set of behaviors than even the combination of the two separate approaches, such as the nonmonotonic behavior in the path of expected values over time.

Financial Models

The financial models in this section are relative value models in that they value one security in relation to one or more other securities. To some extent, the chapters build on one another, with the material for one chapter serving as a starting point for the material in another chapter. And the links between asset, basis, and credit default swaps result in strong interdependencies between the chapters treating them, which therefore together form a “swap block” of chapters.

Some Comments on Yield, Duration, and Convexity

A working knowledge of bond and interest rate mathematics is a prerequisite for this book. But we believe some of the basic bond math taught to practitioners is simply wrong, or at the very least misleading. For example, the *basis point value* of a bond is fundamentally a different concept from the *value of a basis point* for a swap, yet many practitioners are unclear about this difference. As another example, the Macaulay duration of a bond is often referred to as the weighted average time to maturity of a bond, but this is only true when all the zero-coupon bonds that constitute the coupon-paying bond have the same yield, a condition that is almost never observed in practice. We also discuss the frequent misuse of bond convexity and suggest a more practical interpretation of the concept.

Some Comments on Yield Curve Modeling

While yield curve models take center stage in many academic papers, the practical focus of this book downgrades their role to supporting concrete analytic tasks. But even here, they are ubiquitous. This chapter therefore reflects on the implicit model assumptions. It also offers some thoughts about mixed jump-diffusion processes and shadow rate models.

Bond Futures Contracts

A simple no-arbitrage relation applies to the relative values of a cash bond, the repo rate for the bond, and the forward price of the bond. But government bond futures contracts typically contain embedded delivery options, which complicate the analysis. We present a multi-factor model for valuing the embedded delivery option, which can be implemented in a spreadsheet using basic stochastic simulation.

Fitted Bond Curves

Fitting curves to the prices of government bonds observed in the market provides two main results: (1) times series of constant maturity par (or zero) bond yields as input variables for a PCA, for example; and (2) a fair value for every individual bond, i.e. a common yardstick to measure its richness or cheapness relative to other bonds. We discuss different functional forms and add external explanatory variables to the curve fitting process, allowing adjustment for impacts from benchmark status or repo specialness on bond prices.

An Analytic Process for Government Bond Markets

Before starting the "swap block" of chapters, we take stock of the statistical and financial models presented so far and illustrate how they could be combined into a comprehensive analytical process for government bond markets.

Overview of the Following Chapters: Asset Swaps, Basis Swaps, Credit Default Swaps, and Their Mutual Influences

All global asset swaps, basis swaps, and credit default swaps are linked, resulting in interesting and complex relationships. A consequence for the presentation in this book is, therefore, that the individual pricing of the different types of swaps takes place in the framework of mutual dependencies. This framework is sketched at the beginning of the swap block of chapters in order to equip the reader with a map through the multitude of relationships between the instruments described in following chapters. Once the journey through the swap block of chapters is completed, these mutual influences can be analyzed quantitatively to a certain extent.

Reference Rates

The key conceptual approach to price swap spreads is to consider them as a basis swap between the repo rate of the bond and the reference rate of the swap, the types of which have increased following the (partial) transition away from LIBOR. This chapter classifies the different reference rates, provides a model for the spreads between them and for their spread to the repo rate. The basis between unsecured and secured reference rates is identified as a major driving force of these spreads.

Asset Swaps

With this foundation it is straightforward to calculate the impact of different funding rates (repo for the bond, LIBOR, OIS or SOFR for the swap) on asset swap spreads. Since an asset swap not only exchanges funding rates but also credit exposure, the impact of different credit risk needs to be added to the swap spread model. The CDS provides a market price for the credit risk of bonds and thus is an obvious candidate to be included among the input variables for the swap spread model.

Credit Default Swaps

However, the price of sovereign CDS usually expresses the credit risk together with the value of the delivery option and of the fact that in case of default the compensation is paid in USD. The foremost task of this chapter is therefore to extract the pure credit information from the CDS market in order to be able to use it as input in the swap spread formula. Unfortunately, the scarcity of observations of previous relevant government defaults results in a significant estimation error. We then discuss also other applications of CDS in RV analysis and trading.

Intra-Currency Basis Swaps

ICBS exchange two different reference rates in the same currency. Hence, they can be valued by applying the model for spreads between reference rates.

Cross-Currency Basis Swaps

CCBS exchange reference rates in different currencies. Since by convention one leg is always USD SOFR and there is no secured reference rate in other currencies, they also involve an exchange between different types of reference rates. By combining ICBS and CCBS, any reference rate in any currency can be swapped into any other. These basis swaps therefore fulfill the function of building blocks and of linking all global swap markets.

Combinations and Mutual Influences of Asset, Basis, and Credit Default Swaps

By combining asset and basis swaps, every bond can be expressed as a spread versus USD SOFR, which is therefore a universal yardstick for comparing all global bonds and the basis for asset allocation and funding decisions. Likewise, the CDS also measures every bond as a spread versus USD SOFR. Using the insights into the delivery option and USD-redemption from the CDS chapter, we present arbitrage relationships between both.

The existence of a universal yardstick and of arbitrage relationships results in strong interdependencies between all asset, basis, and credit default swap markets worldwide. Their analysis is therefore a complex task which should take all these mutual influences into account. We present case studies to illustrate how the links work in practice.

Global Bond RV Via Fitted Curves and Via SOFR Asset Swap Spreads

We show that the deficiencies of swap spreads as rich/cheap indicator for bonds are a consequence of the difference between the swap and bond yield curves, which are manageable in case of SOFR swap spreads only. Hence, to compare bonds in USD, the universal yardstick "SOFR swap spreads" provides a suitable alternative to fitted curves. When comparing bonds in another currency, however, fitting a curve through the basis swapped bonds remains the only reasonable option.

Other Factors Affecting Swap Spreads

At the end of the swap block of chapters we consider those driving factors not captured in its conceptual foundation, such as the impact of increasing leverage constraints and shadow costs. Since some of these variables are unobservable, the goal of modeling and predicting the relationships between bonds and swaps allows only partial and approximative solutions.

Options

We address the analysis and trading of options in a relative value context by discussing three broad categories of option trades. In the first, the trader simply buys or sells an option with a view that the underlying will finish in-the-money or out-of-the-money, with no dynamic trading. In the second, the trader attempts to capitalize on the difference between the implied volatility of the option and the actual volatility that the trader anticipates for the underlying instrument, by trading the option against a dynamic position in the underlying. In the third, the trader positions for a change in the implied volatility of the option, irrespective of the actual volatility of the underlying instrument.

Relative Value in a Broader Perspective

We conclude our sometimes rather technical description of relative value analysis by taking a broader perspective on its macroeconomic functions. At a time when professionals in the financial services industry increasingly need to justify their role in society, we present a few thoughts about the benefits of arbitrage for society.

Throughout the book, we offer pieces of general advice – words of wisdom that we've gleaned over time. We've been mentored by some of the best in the business over the years, with particular thanks to our managers and colleagues in Anshu Jain's Global Relative Value Group at Deutsche Morgan Grenfell, and especially to David Knott, Pam Moulton, and Henry Ritchotte. They were good enough to impart their wisdom to us, and we're happy to pass along this treasure trove of useful advice, hopefully with a few additional pearls of insight and experience that we've been able to add over the years.[1]

Please visit the website accompanying this book to gain access to additional material, www.wiley.com/go/fixedincome

[1]When reviewing the first edition of this book, Christian Carrillo, Martin Hohensee, Antti Ilmanen, and Kaare Simonsen provided valuable feedback, enhancing our product. Many participants of our training courses have contributed to the improvements of the second edition.

PART

I

Statistical Models

CHAPTER 2

Mean Reversion

WHAT IS MEAN REVERSION AND HOW DOES IT HELP US?

Mean reversion is one of the most fundamental concepts underpinning relative value analysis. But while mean reversion is widely understood at an intuitive level, surprisingly few analysts are familiar with the specific tools available for characterizing mean-reverting processes.

In this chapter, we discuss some of the key characteristics of mean-reverting processes and the mean reversion tools that can be used to identify attractive trading opportunities. In particular, we address:

- model selection;
- model estimation;
- calculating conditional expectations and probabilities;
- calculating ex ante risk-adjusted returns, particularly Sharpe ratios;
- calculating first passage times, also known as stopping times.

For each concept, we start with a verbal and intuitive description of the concept, followed by a mathematical definition of the concept, and finish with an example application of the concept to market data.

A variable is said to exhibit mean reversion if it shows a tendency to return to its long-term average over time. Mathematicians will object that this definition is simply an exercise in replacing the words "exhibit," "mean," and "reversion" with the synonyms "shows," "long-term average," and "return". To address such objections, we'll provide a more mathematical definition shortly. But first we'll attempt to establish some further intuition about mean-reverting processes. To some extent, Justice Stewart's famous maxim on pornography, "I know it when I see it," applies to mean reversion. With that in mind, let's take a look at some processes that exhibit mean reversion and a few that don't.

Figure 2.1 and Figure 2.2 show two simulated time series. Both have an initial value of zero, and both have identical volatilities. But one is constructed to be a simple random walk, with zero drift, while the other is constructed to have a tendency to return toward its long-run mean, constructed to be zero